

MOUNTAINS AND VOLCANOES

Most mountains are part of chains that formed where two or more plates of Earth’s crust collided. A nearly continuous range of mountains (the Circum-Pacific System) runs for 25,000 miles (40,000km) around most of the Pacific Basin and includes three-quarters of the world’s active and dormant volcanoes.

Its mountains include the Andes of South America, the Western Cordillera of North America, and the Japanese Alps. A second, almost unbroken mountain belt (the Alpine–Himalayan System) runs from Morocco through Europe and southern Asia to Southeast Asia, including Earth’s greatest mountain range, the Himalayas.

North America

Mount Katmai

Location Alaska, northwest U.S.

This snow-covered volcano is about 6 miles (10km) in diameter, with a massive, lake-filled caldera at its center. It was once thought to be responsible for the 20th century’s biggest eruption, in June 1912, but scientists later showed that the nearby Novarupta lava dome was the actual source. When Novarupta erupted, a magma chamber beneath Mount Katmai emptied, collapsing its summit and creating the caldera. Thousands of steam vents peppered the area for years after the eruption, inspiring the name Land of Ten Thousand Smokes.

The Mackenzie Dikes

Location From the Great Lakes to the Arctic, western Canada

The Mackenzie Dikes are Earth’s largest dike swarm—a large group of near-vertical, sheetlike formations of igneous rock. The dikes reach the surface across a vast area that is about 1,900 miles (3,000km) by 310 miles (500km). Mostly composed of basalt, they

were intruded into the ancient rocks of the Canadian Shield about 1.3 billion years ago.

The Palisades

Location Along part of the western shore of the Hudson River, eastern U.S.

The Palisades are dramatic, almost sheer cliffs that stretch 20 miles (32km) along the west side of the Hudson River (see p.363) from near Jersey City to Nyack. The cliffs range in height from 295 to 540 ft (90 to 165m) and are the eroded edge of

the Palisades Sill, a large mass of an igneous rock called diabase. The sill is about 200 million years old.

The Sierra Nevada

Location Mainly in California, western U.S.

The magnificent skyline and dramatic, glacier-eroded valleys of the Sierra Nevada range make it one of the most beautiful regions of the U.S. Fast-flowing rivers, valleys cloaked in pine forest, mountain meadows, and granite peaks are all

characteristic of the area. One of the Sierra Nevada’s peaks, Mount Whitney, is, at 14,505 ft (4,421 m), the highest in the lower 48 states. The range extends about 400 miles (640km) from south to north and 70 miles (110km) west to east.

The Devil’s Postpile

Location The Sierra Nevada range, California, western U.S.

The Devil’s Postpile National Monument is situated on the western slope of the Sierra



► FALL FOLIAGE AT THE PALISADES SILL